Sahaj Bhadja

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Summary

Software Engineer pursuing an MSCS at USC with 2+ years building and refactoring scalable, high-query-volume databases and backend services in Python, SQL, MongoDB, and distributed systems, with hands-on experience in parallel processing, data pipelines, and performance debugging.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Created and refactored **relational and NoSQL databases** to withstand high query volumes for 500+ users, optimizing indexing and query patterns to cut latency from 1.2s to 350ms.
- Designed **parallel async orchestration** with explicit job-state transitions and failure isolation, building scalable data pipelines that processed 5K+ monthly jobs at 99%+ reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Built and enhanced **Python backend features** with MySQL and REST APIs for 150+ users, writing, testing, and debugging issues across development, test, and production environments.
- Engineered **parallel batch processing** (multithreading) handling 10K+ daily requests, debugging performance bottlenecks to sustain stable response times under peak load.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Designed scalable **microservices and APIs** with OOP patterns and Mockito testing, debugging performance issues to improve response times **30%** and reduce technical debt **20%**.
- Built **ETL data pipelines** (AWS Glue, S3) with nightly automated testing across three operating systems, boosting pipeline efficiency **68%** in an Agile environment.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed a **microservice platform** with MongoDB and REST APIs across eight services, syncing 2M+ records with real-time WebSocket updates and JWT role-based access.
- Built **real-time monitoring** (Sentry, Kibana) and automated Docker/Jenkins CI/CD on Azure, cutting deployment time **50%** and improving scalability through modular service design.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built **distributed data pipelines** (Databricks, Spark) and Flask APIs for real-time analytics on 50K+ records, delivering sub-200ms responses.
- Integrated payment and webhook APIs with token-based auth and Firebase real-time sync, reducing manual handling **70%** and improving cross-platform performance.

Skills

Languages: Python, SQL, Java, JavaScript/TypeScript, C++, Bash
Backend & APIs: Django, Flask, FastAPI, Node.js, REST APIs, RPC, Microservices
Databases: MongoDB, MySQL, PostgreSQL, Redis, SQL Server, Azure SQL
Data & Messaging: ETL Pipelines, Data Modeling, Metadata Extraction, Message Queues, Hadoop, Spark
Distributed & Performance: Parallel Programming, Concurrency, Scalability, Query Optimization, Caching, Performance Tuning
Cloud & DevOps: AWS, Azure, Docker, Kubernetes, CI/CD, Git, Monitoring

Projects

Big-Brain Live | *Python, GCP, PostgreSQL, FAISS*
- Built a **platform for data storage and metadata extraction** on GCP processing 50K+ records with semantic indexing, incremental updates, and tuning that cut retrieval latency **40%**.

FinNexus Live | *Kubernetes, MongoDB, WebSockets*

- Built a **Kubernetes-based platform** with MongoDB supporting 50K+ users and 1M+ transactions, with real-time updates, token-based access, and scalable CI/CD deployment.

PiXMod Live | *MongoDB, CLIP, Transformers, Next.js*

- Engineered **MongoDB metadata indexing** and scalable image storage with CLIP embeddings across 100K+ vectors, powering semantic search in an AI SaaS platform.